BNO085 Embedded Heading Sensor System

STM32F767ZI + FreeRTOS + NMEA 0183 + ROS2 lifecycle driver

Objective

- Acquire calibrated magnetic-field data from BNO085
- Calculate magnetic heading and reduce fluctuation on STM32
- Transmit standard NMEA HDM sentences over UART
- Publish ROS2 IMU orientation, TF and diagnostics

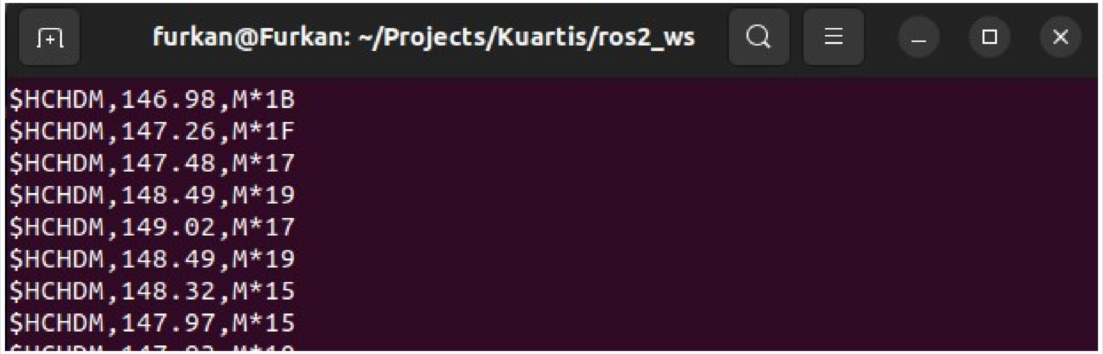
Deliverable focus: reliable sensor acquisition, timing validation and ROS2 integration.

End-to-End System Architecture

From BNO085 calibrated magnetic field to ROS2 orientation and diagnostics.

BNO085 magnetometer → **STM32 SPI/SHTP** → **Calibration + heading** → **Kalman X/Y filters** → **NMEA HDM UART** → **ROS2 node**

NMEA output



$HCHDM,<heading>,M*checksum

Key interfaces

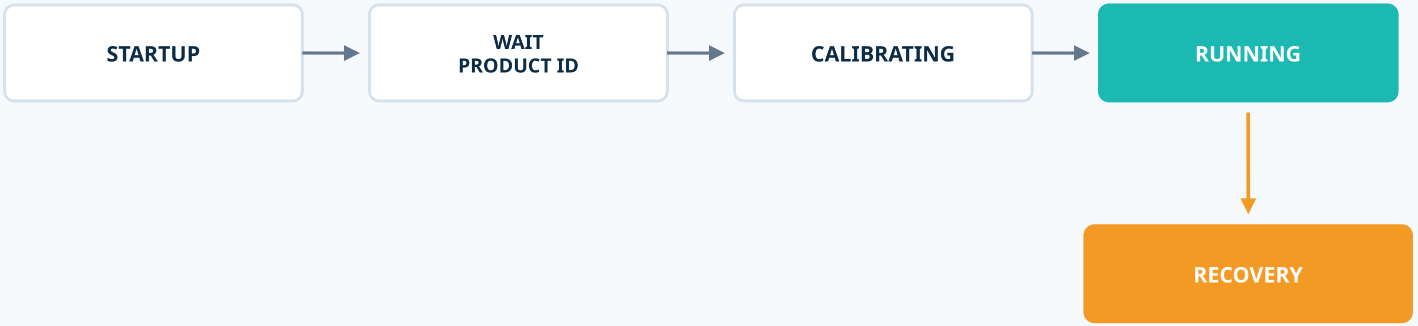
- BNO085 communication through SPI and SHTP packets
- UART stream uses readable NMEA 0183 HDM format
- ROS2 converts heading into yaw quaternion and TF

Embedded Runtime Architecture

FreeRTOS tasks isolate sensor acquisition from communication output.

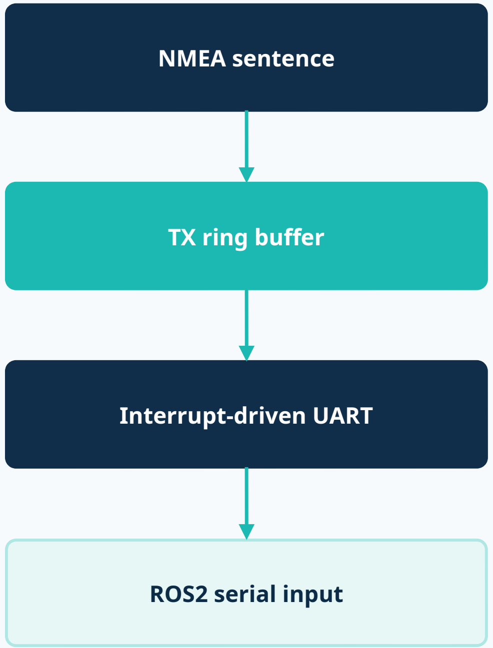
Task-level design

Sensor state machine



Timeouts and communication errors trigger recovery and re-initialization.

UART output path



Complete messages are enqueued to avoid partial NMEA frames.

Heading Calculation & Kalman Filtering

Heading is calculated from calibrated X/Y magnetic components and stabilized before ROS output.

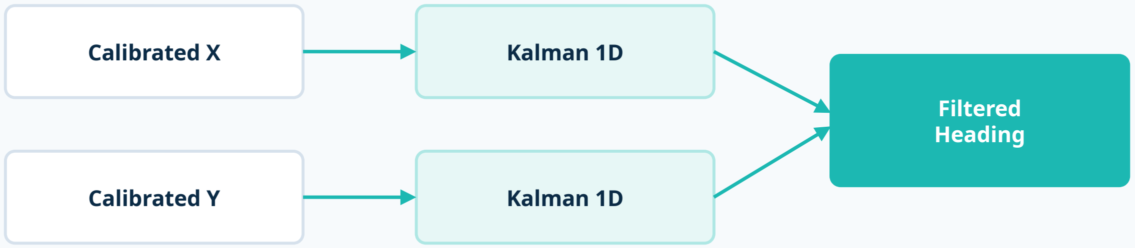
Heading calculation

$$\text{heading} = \operatorname{atan2}(Y, X) \times 180 / \pi$$

if heading < 0° → heading = heading + 360°

The output is a magnetic heading in the 0° to 360° range.

Filtering strategy



Kalman parameter selection

Measurement noise R

Stationary magnetic-field variance was estimated with Welford algorithm.

$R_x = 0.433\ \mu T^2$ · $R_y = 0.393\ \mu T^2$

Process noise $Q = 0.25\ \mu T^2/s$ was tuned for smoothness vs responsiveness.

Core update equations

$$K = P^- / (P^- + R)$$

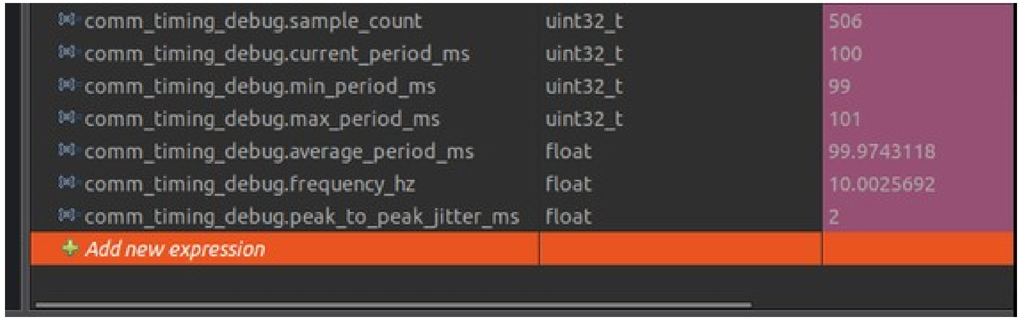
$$x = x^- + K(z - x^-)$$

Kalman gain is recalculated at every update.

Performance Validation

Embedded timing and ROS2 publish rate were measured separately.

Embedded timing counters



Embedded result: 99-101 ms period, 2 ms peak-to-peak jitter, 10.003 Hz.

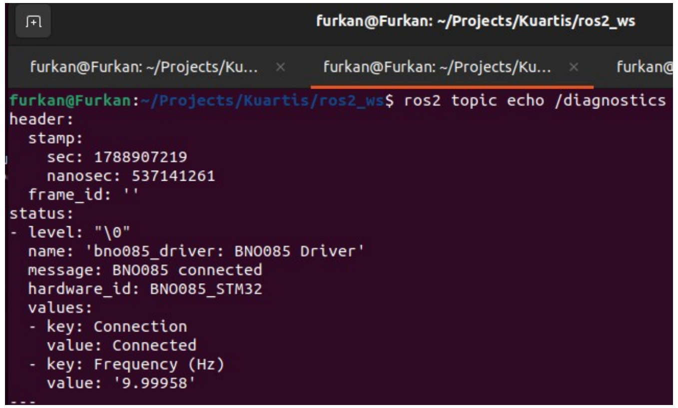
ROS2 publish rate

ROS2 result: /imu/data remains at approximately 10 Hz over 849 samples.

ROS2 Integration Outputs

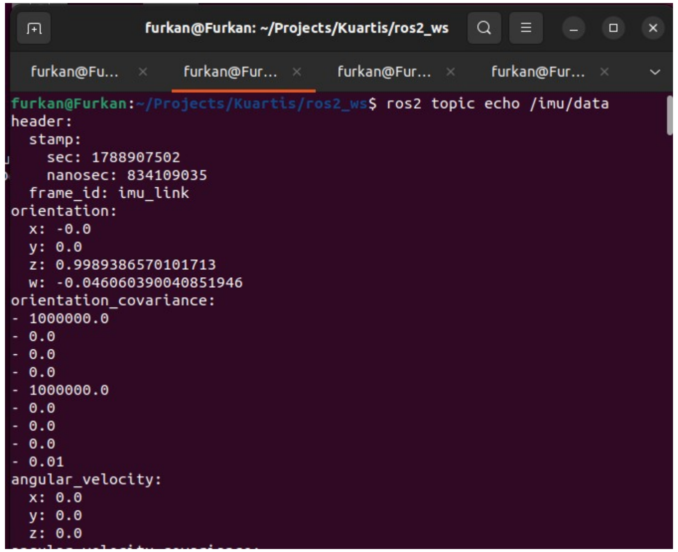
Lifecycle node publishes standard ROS interfaces and runtime health information.

Diagnostics output



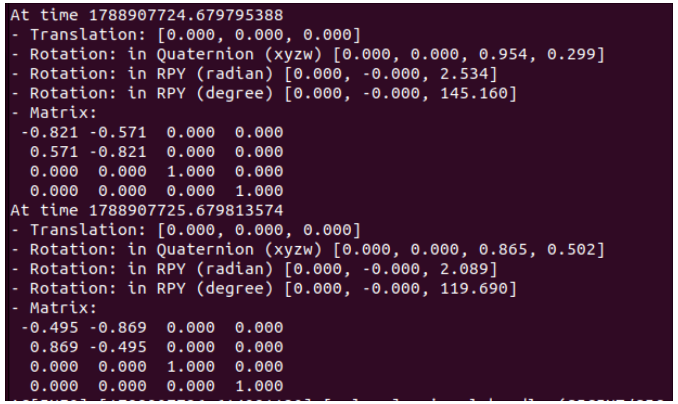
Connection: Connected, frequency: ~10 Hz.

/imu/data message



Heading is represented as a yaw quaternion.

TF output



base_link to imu_link transform exposes yaw.

Challenges, Improvements & Live Demonstration

The final step validates the complete embedded-to-ROS data path live.

Challenge

Magnetometer data still showed short-term heading fluctuation after calibration.

Cause: measurement noise and environmental magnetic disturbance.

Solution

Apply independent scalar Kalman filters to calibrated X and Y before heading calculation.

R was measured from stationary variance; Q was tuned experimentally.

Improvement

Add tilt compensation to improve heading reliability when the sensor is not level.

A future version can fuse accelerometer/orientation data for compensated heading.

Live demonstration sequence

1. **CubeIDE debug**
   Check calibration_ok and runtime variables.

2. **CubeMonitor graph**
   Compare raw heading and Kalman-filtered heading.

3. **ROS2 topics**
   Show /imu/data content and publish rate.

4. **TF + diagnostics**
   Show yaw transform and connected status.